FINTECH & BANKING ENGINEERING SQUAD

SOMBANK TP V5

.NET Framework Architecture • Angular Single-Page Apps • Banking Change Requests & Zero Regression

Client: Sadak Samatar, CTO — SomBank
Platform: Transaction Platform V5 (TP V5 — 2-Year Ongoing Engine)
Capacity: 2 Dedicated Full-Stack Engineers + Project Manager
Commitment: Contract-to-Hire Available • Strict NDA Ready

TP V5 Change Request Execution: Precision .NET & Angular Banking Engineering

A senior, battle-tested FinTech engineering squad ready to assume ownership of SomBank's confidential Change Request (CR) backlog for the flagship Transaction Platform (TP V5). Combining deep .NET Framework backend architecture, responsive Angular client applications, immutable audit logging, and zero-regression verification to safeguard live banking operations.

Backend: **.NET Framework • C# • Web API • SQL Server**
Frontend: **Angular (SPA) • TypeScript • HTML5/CSS3**
Domain: **FinTech • Core Banking • Transaction Integrity**
Security: **OAuth2 / JWT • OWASP • Strict Confidentiality**
Delivery: **Fast Impact Analysis • Automated Regression Testing**

1. EXECUTIVE SUMMARY & DELIVERY APPROACH

Engineered for a Mature 2-Year Production Banking Core

Managing Change Requests on Mature Banking Systems

RISK AWARENESS

A banking platform running continuously for nearly two years under CTO leadership represents mature, mission-critical infrastructure where even minor modifications carry operational risk:

- **Zero-Regression Mandate:** New feature requests and workflow alterations must never disrupt existing customer balance calculations, payment clearing, or third-party API hooks.
- **Confidentiality & Compliance:** SomBank's change specifications contain sensitive financial rules and institutional workflows requiring strict NDA protocol and isolated testing.
- **Handoff Continuity:** Smoothly bridging the transition from the former CTO account to CTO Sadak Samatar with clean git history, documented code, and architectural consistency.

The Dedicated Squad Advantage for SomBank

TWO-ENGINEER CAPACITY

SomBank is seeking two engineers to accelerate TP V5 change delivery. Our squad delivers the exact dual-engineer throughput plus project governance:

- **Senior .NET & Backend Architect (Viranga Lakshan):** Manages C# .NET Web APIs, transaction orchestration, database integrity, and banking security protocols.
- **Angular & Frontend/Database Developer (Janith Kavishka):** Specializes in Angular component architecture, responsive financial dashboards, and SQL Server transaction tracking.
- **FinTech Delivery Project Manager (Lithumitha Ravikumar):** Oversees CR traceability, QA acceptance test validation, and zero-regression signoff.

2. TECHNICAL SCOPE & CAPABILITIES ALIGNMENT MATRIX

Direct Alignment with SomBank TP V5 Requirements

SOMBANK TP V5 REQUIREMENT	SQUAD IMPLEMENTATION & ENGINEERING STANDARD	VERIFIED PRODUCTION PRECEDENTS
.NET Framework Backend Architecture	C# / .NET Web API development, Entity Framework / Dapper database layers, business logic encapsulation, and asynchronous task execution.	Architected enterprise C# / .NET components, RESTful API gateways, and multi-service business domain logic.
Angular Single-Page Application (SPA)	Modern Angular architecture, RxJS reactive state management, modular components, TypeScript typing, and CSS/HTML responsive financial UI.	Built complex financial tracking dashboards, approval matrices, and customer/admin portals with real-time UI states.
FinTech & Transaction Integrity	Double-entry accounting models, idempotency keys, ACID-compliant database transactions, ledger reconciliation, and tamper-evident audit trails.	Engineered nationwide financial field collection & receipting for SINGER PLC and 3-way reconciliation engine (*InvoiceIQ*).
Confidential Change Request (CR) Protocol	Formal impact analysis before modifying code: dependency mapping, database schema regression checks, and isolated staging validation.	Enterprise change management on multi-tenant SaaS commerce platforms and multi-squad delivery governance.
Banking Security & Access Control	Role-Based Access Control (RBAC), OAuth2 / JWT authentication, data-at-rest & in-transit encryption, and OWASP Top 10 hardening.	Implemented Keycloak OAuth2/OIDC token verification middleware, secure S3 pipelines, and ClamAV antivirus scanning.

Immediate Availability & Confidential NDA Execution

We are prepared to sign SomBank's Non-Disclosure Agreement (NDA) immediately upon Upwork contract initiation. Our squad is fully available for long-term engagement and contract-to-hire consideration under CTO Sadak Samatar's direct supervision.

FULL-STACK .NET & ANGULAR BANKING ENGINE

ARCHITECTURE

Stack: .NET Framework + Angular (SPA)
Database: SQL Server / ACID Compliant
Methodology: Zero-Regression Impact Analysis

Robust Engineering for High-Throughput Financial Systems

3. FULL-STACK BANKING ARCHITECTURE (.NET FRAMEWORK + ANGULAR)

Enterprise Modularity Designed for Maintainability and Scale

Backend: .NET Framework & Secure APIs

C# / WEB API

We write clean, idiomatic C# .NET code following enterprise Domain-Driven Design (DDD) principles:

- **Service & Repository Layering:** Strict decoupling of business domain rules from data persistence and controller endpoints, ensuring changes in logic do not corrupt persistence.
- **Defensive Transaction Processing:** Wrapping multi-step banking operations in strict database transactions with distributed locking and rollback safeguards.
- **High-Performance Data Access:** Optimizing Entity Framework / ADO.NET / Dapper queries, avoiding N+1 bottlenecks, and enforcing database query execution time limits.

Frontend: Angular Component Architecture

TYPESCRIPT / CSS

Modern, maintainable Angular client engineering for seamless banking portal user experiences:

- **Reactive State Management:** Clean RxJS observables, behavior subjects, and centralized state stores preventing race conditions in financial forms.
- **Modular Feature Modules:** Lazy-loaded modules for transactions, approvals, customer profiles, and reports ensuring fast initial load times.
- **Strict Validation & UI Guardrails:** Real-time form validation, currency masking, decimal precision formatting, and double-click prevention on payment execution buttons.

4. CONFIDENTIAL CHANGE REQUEST (CR) EXECUTION PROTOCOL

4-Step Discipline to Ensure Zero Production Regression

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[STEP 1: IMPACT AUDIT] ⟶ [STEP 2: SANDBOX CODING] ⟶ [STEP 3: REGRESSION QA] ⟶ [STEP 4: CTO REVIEW] • Review Confidential CR • Branch-isolated feature • Automated Unit Tests • Pull Request Walkthrough • Map dependent services • Clean C# / Angular code • Ledger & Balance Checks • Zero-Downtime Deployment • Identify Schema Changes • Zero hardcoded configs • Cross-browser UI verify • Post-launch verification
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Impact Analysis & Schema Review

STEP 1 & 2

Before touching existing production code, we review the confidential CR document and generate a technical impact matrix: identifying affected stored procedures, API contracts, and Angular components to prevent unintended side effects.

Automated Testing & Release Governance

STEP 3 & 4

Every change is validated against regression test suites verifying that existing transaction workflows, balance updates, and export reports function identically. Code is delivered via atomic git commits with clear pull request summaries.

5. BANKING SECURITY, AUDIT LOGGING & COMPLIANCE

Protecting Sensitive Somali Banking Data & Transactions

Immutable Audit Trails

AUDITABILITY

Every state-changing operation records who, what, when, IP address, and previous state into tamper-evident database audit tables for regulatory compliance.

OWASP Hardening

APP SECURITY

Protection against SQL Injection, Cross-Site Scripting (XSS), CSRF token validation, input sanitization, and secure HTTP-only cookie headers.

Least-Privilege RBAC

ACCESS CONTROL

Granular permissions mapping ensuring branch staff, compliance officers, and system administrators only access authorized financial workflows.

VERIFIED FINTECH & ENTERPRISE EXPERIENCE

Production Banking, Payment Processing, and Enterprise .NET Systems

TRACK RECORD

Domain Precedents: FinTech & Financial Operations
Backend Tech: .NET Framework, C#, Java/Spring, SQL
Frontend Tech: Angular, React, TypeScript

6. ENTERPRISE PRODUCTION CASE STUDIES

Real Financial and Enterprise Systems Delivered by Squad Members

Enterprise Financial Field Collection & Receipting Platform

Client: SINGER Sri Lanka PLC (DMS Electronics) • Janith Kavishka

FINANCIAL WORKFLOWS

Mission-Critical Financial Platform: Designed and developed a field financial collection and receipt management platform handling batch-wise transaction tracking, mobile receipt generation, and real-time reconciliation for nationwide retail and consumer finance.

- **Database Integrity & Migration:** Executed complete zero-downtime database migration to Amazon RDS (MySQL) with zero loss across millions of financial records.
- **Message-Driven Event Streaming:** Implemented RabbitMQ message queues to guarantee asynchronous, decoupled processing of high-volume financial transactions.
- **Continuous Production Operations:** Provided ongoing system design and production operational support ensuring uninterrupted uptime for island-wide branch operations.

DIMO Enterprise Gate Pass & Logistics Management System

Client: DIMO Lanka (Enterprise Logistics) • Viranga Lakshan

SECURITY & SAP ERP

High-Security Enterprise Platform: Architected API gateway and core domain services managing vehicle gate pass dispatch, SAP ERP financial integration, and multi-role approval hierarchies.

- **Keycloak OAuth2 / JWT Security:** Handled centralized authentication, token verification middleware, and granular role-based authorization for multi-level managers.
- **Event-Driven Audit Trails:** Integrated Spring Kafka publishing immutable audit events for all approvals, rejections, and material movements.
- **Secure File Handling with ClamAV:** Engineered AWS S3 upload pipeline with automated asynchronous virus scanning and quarantine buckets.

Enterprise C# & .NET Integrations

Al Ain Museum & Holsten Arabia

.NET & C#

Developed and customized C# / .NET backend components, Sitecore Helix architecture, and RESTful service integrations supporting enterprise content and transaction delivery.

Guardian Funds Financial Dashboard

Client: Guardian Funds (Australia)

FINTECH PM

Managed delivery of investment portfolio analytics dashboard and corporate investment platform. Coordinated QA test validation against strict financial acceptance criteria.

EXECUTION ROADMAP & SQUAD PROFILES

DELIVERY

Budget Alignment: $5,796 Milestone Scope
Model: 2 Dedicated Engineers + PM
Availability: Immediate Start • Full-Time Overlap

Structured Phasing, Team Expertise, and Immediate Onboarding for SomBank

7. PHASED CHANGE REQUEST IMPLEMENTATION PLAN

Systematic Execution from Confidential Document Ingestion to Deployment

Phase 1: Confidential Ingestion & Architecture Audit

DAYS 1–3

- Execute Upwork contract and sign SomBank Non-Disclosure Agreement (NDA).
- Receive confidential TP V5 change request documentation from CTO Sadak Samatar.
- Review existing .NET Framework solution structure, Angular SPA repo, and SQL Server schema.
- Deliver technical Change Impact Matrix and milestone breakdown for CTO signoff.

Phase 2: .NET Backend & API Implementation

SPRINT 1

- Implement backend C# / .NET changes: updated endpoints, business logic services, and models.
- Execute safe database schema migrations, stored procedure updates, and index tuning.
- Enforce transaction isolation, parameter validation, and comprehensive unit tests.
- Provide staging API endpoints with Swagger/Postman documentation for frontend consumption.

Phase 3: Angular Frontend & UI/UX Enhancements

SPRINT 2

- Build and refactor Angular components, templates (HTML5), and styling (CSS3/SCSS).
- Integrate updated .NET REST APIs with robust error handling and loading indicators.
- Implement client-side field validation, currency masks, and role-based view controls.
- Verify full responsiveness across desktop, tablet, and mobile browsers.

Phase 4: Regression QA, Handover & Support

SPRINT 3

- Execute end-to-end regression testing across all existing and newly modified workflows.
- Conduct live pull request code review with CTO Sadak Samatar and SomBank tech team.
- Deploy changes to UAT/Staging, support user acceptance testing, and assist production deployment.
- Provide 30 days of warranty support and prepare for ongoing contract-to-hire extension.

8. MEET THE SOMBANK ENGINEERING SQUAD

Senior FinTech Specialists Dedicated to TP V5 Success

Viranga Lakshan

LEAD FULL-STACK / .NET ARCHITECT

Associate Software Engineer, Arimac • 4+ Years Enterprise Systems

Leads backend systems architecture, C# / .NET Web API engineering, banking security protocols, OAuth2/JWT integration, and database transactional integrity.

.NET Framework | C# / Web API | SQL Server | OAuth2 / JWT

Janith Kavishka

ANGULAR & DATABASE ENGINEER

Programmer, DMS Electronics • Financial Workflows & Full-Stack

Specializes in Angular SPA development, TypeScript, responsive CSS/HTML UI, financial transaction tracking, and database query optimization.

Angular (SPA) | TypeScript | CSS3 / HTML5 | FinTech Tracking

Lithumitha Ravikumar

PROJECT MANAGER

Associate Project Manager, Arimac • Enterprise SaaS & Financial Platforms

Manages Change Request traceability, sprint coordination, QA acceptance testing against banking specifications, and technical delivery documentation.

Project Management | CR Traceability | QA Acceptance | Release Governance

Commercial Alignment with SomBank's $5,796 Budget

Our squad is fully aligned with SomBank's fixed-price budget of **$5,796.00**. We are structured to provide the capacity of **two dedicated developers** collaborating seamlessly under one contract, with long-term availability for contract-to-hire extension under CTO Sadak Samatar.